PROJECT 6 — Exoplanet Transit Detection (Kepler)

Domain	Exoplanet science — detection of planetary transits in stellar photometry
Hardware	Laptop only. lightkurve, astropy, scikit-learn, and Keras or PyTorch for the 1D convolutional model.
Dataset	NASA Kepler mission data: the Kepler Objects of Interest catalogue with vetted dispositions, and the associated stellar light curves retrievable through the lightkurve package.
ML Tool	A 1D convolutional network on folded light-curve views, with a scikit-learn model on catalogue features as the comparison
Indicative Cost	Nil
Difficulty	Medium-high — the most involved preprocessing in this document, and the dataset contains a leakage trap that catches most teams on their first attempt

6.1 Problem Statement & Motivation

When a planet passes in front of its star, the star's measured brightness dips by a fraction of a percent for a few hours, and repeats every orbital period. Kepler stared at a single patch of sky for years to catch exactly this signal, and the resulting archive is one of the great open datasets in astronomy. The difficulty is that most things that look like a transit are not planets: eclipsing binary stars produce far deeper but superficially similar dips, and the spacecraft itself introduces systematic artefacts. Human vetters spent enormous effort separating genuine planet candidates from these false positives, and their verdicts are recorded in the catalogue. This project asks teams to reproduce that judgement automatically. It also contains a deliberate trap, described below, that teaches the most important lesson in applied machine learning more memorably than any lecture can.

6.2 Objectives

1. Retrieve the Kepler Objects of Interest catalogue and a working subset of light curves, recording the data release and the exact query used.
2. Conduct a leakage audit of the catalogue columns and remove every field that encodes or summarises the vetting verdict itself, documenting which columns were removed and why.
3. Detrend the light curves, fold them on the reported orbital period, and construct binned global and local views of the folded signal.
4. Construct a split grouped by host star, and assert programmatically that no star contributes rows to more than one side of the split.
5. Train a 1D convolutional network on the views, and a scikit-learn model on the surviving catalogue features, on identical splits.
6. Evaluate with precision-recall analysis rather than accuracy, and inspect the false positives to establish what the model has learned to reject.

6.3 Design Parameters

Parameter	Target / Specification	Rationale
Dataset	Kepler Objects of Interest cumulative catalogue plus light curves via lightkurve; release and access date recorded	The catalogue is periodically revised as candidates are re-vetted
Label definition	Confirmed planets and validated candidates versus vetted false positives; the exact disposition mapping stated in writing	Different reasonable mappings exist and the choice must be deliberate rather than inherited
LEAKAGE AUDIT	Any column carrying a vetting score, a pipeline verdict, or a flag derived from the disposition must be removed before	This is the central trap of the dataset. A model trained with these columns present reports near-

Parameter	Target / Specification	Rationale
	modelling. The removed list appears in the report.	perfect performance and has learned nothing — it is reading the answer.
Detrending	Remove stellar variability and instrumental trends with a stated method and window; the same treatment applied to every light curve	Untreated stellar variability swamps a transit of a few hundred parts per million
Folding and views	Fold on the reported period; construct a wide global view of the whole orbit and a narrow local view zoomed on the transit	This two-view representation is the established approach and lets one network see both the shape and the context of the dip
Splitting	Grouped by host star identifier, never by individual row or by cadence	One star can generate several catalogue entries; splitting by row places the same star on both sides
Class balance	False positives outnumber confirmed planets substantially; handled by class weighting and reported	Determines why precision-recall analysis is required
Models	A 1D convolutional network on the views, plus a Random Forest or gradient-boosted model on the audited catalogue features	The tabular model is a strong comparison and often surprisingly competitive
Metrics	Precision-recall curve, area under it, and precision at a fixed high recall; accuracy is not accepted	Follow-up telescope time is the scarce resource, so precision at high recall is the operationally meaningful number
Success criteria	Area under the precision-recall curve ≥ 0.80 after the leakage audit, with recall ≥ 0.90 at a stated precision	Deliberately set post-audit; pre-audit numbers are meaningless

6.4 System / Pipeline Block Diagram

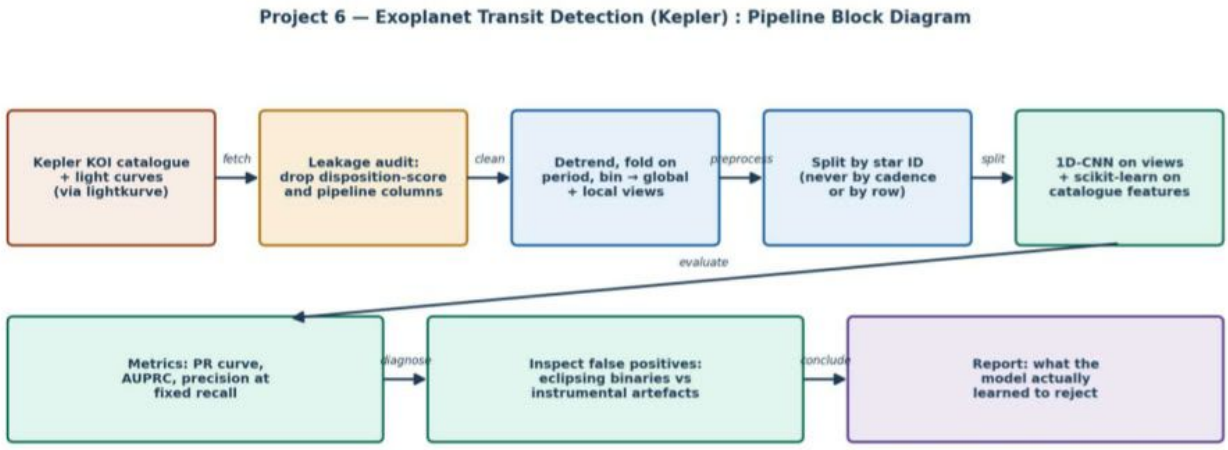


Figure 6.1 — Project 6 pipeline: retrieval, leakage audit and light-curve preprocessing (top), feeding modelling, precision-recall evaluation and false-positive diagnosis (bottom).

6.5 Workflow Flowchart

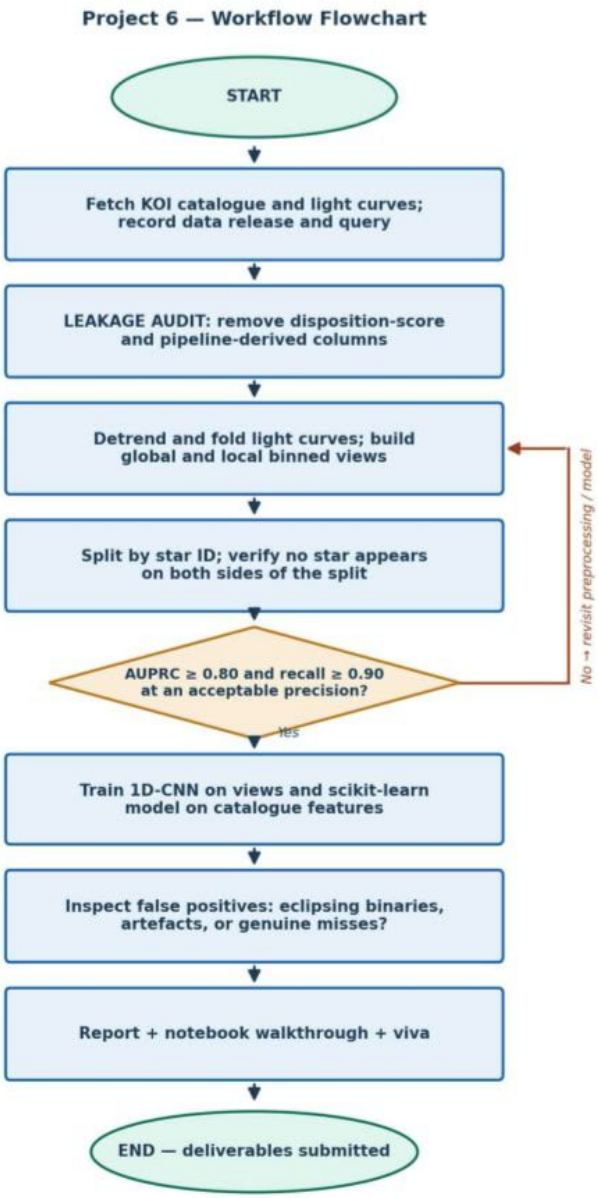


Figure 6.2 — Project 6 workflow. The leakage audit is a mandatory step placed immediately after retrieval, before any modelling decision is taken.

6.6 ML Pipeline (ready-made tools only)

- Retrieval: the catalogue query and the list of downloaded light curves are saved to disk, so the pipeline reruns without re-hitting the archive.
- Leakage audit: teams produce a table of every catalogue column, its meaning, and a keep-or-drop decision with justification. This table is a required report figure and is checked in the viva.
- Light-curve preprocessing: lightkurve handles flattening and folding; the binned global and local views are cached as arrays so that training does not repeat the slow work.
- Split construction: grouped splitting on the host star identifier, with a programmatic assertion that the star sets are disjoint. The assertion appears in the notebook.
- Modelling: a modest 1D convolutional network with two input branches for the two views, plus the tabular comparison model on identical splits with identical class weighting.

- Evaluation and diagnosis: precision-recall curves with bootstrap confidence bands, followed by inspection of the highest-confidence false positives — teams should be able to say whether the model is being fooled by eclipsing binaries, by systematics, or by something else.

6.7 Expected Outcomes & Deliverables

- A reproducible pipeline from cached archive data to predictions.
- A data card recording the catalogue release, the disposition mapping, and the NASA archive acknowledgement.
- The leakage audit table, with a before-and-after performance comparison quantifying exactly how much the leaked columns were inflating the result.
- Precision-recall results for both models with confidence bands, and a stated operating point justified by follow-up cost.
- A false-positive analysis identifying the dominant confusing population.

6.8 Common Pitfalls

- Skipping the leakage audit and reporting a near-perfect score. Teams that do this are asked to redo the evaluation; the inflated number earns no credit.
- Splitting by catalogue row, which lets the same star appear on both sides of the split.
- Folding on a period without checking it, so that a mis-stated period produces a smeared view carrying no transit signal.
- Reporting accuracy, which is dominated by the majority class and conveys nothing about the operational usefulness of the model.
- Describing model outputs as detected planets. The model reproduces a vetting judgement; confirmation requires follow-up observation, and the report must be explicit about that distinction.